Course Code	Course name	L	T	P	C
MATH1065	Advanced Engineering Mathematics – 2	3	1	0	4
Total Units to be Covered: 5	Total Contact Hours: 60				
Prerequisite(s):	Advanced Engineering Mathematics – 1 - MATH1059	Syllabus version: 1.0			

Course Objectives

The aim of the course is to prepare the students to understand and appreciate the power of mathematics as a unifying language transcending a variety of engineering and science disciplines. The focus for the designing of the syllabus has been to provide the students with insights into the mathematical concepts and their applications without much compromising mathematical rigor.

Course Outcomes

On completion of this course, the students will be able to

CO1. Formulate appropriate numerical and optimization schemes for the development of computational algorithms in science and engineering.

CO2. Understand the fundamental tools of complex analysis for addressing and solving a variety of problems viz. special functions, integral transforms, and PDEs.

CO3. Identify and illustrate the use of Bessel functions and Legendre polynomials in real world application.

CO4. Demonstrate computational implementation of integral transforms and their applications.

CO5. Model real-world phenomena evolving in space and time governed by linear PDEs.

CO-PO Mapping

Program Outcomes / Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 1	3	3	2	-	2	-	-	-	-	-	-	-	-	-	-

CO 2	3	3	2	-	1	-	-	-	-	-	-	-	-	-	-
CO 3	3	3	2	-	1	-	-	-	-	-	-	-	-	-	-
CO4	3	3	3	-	3	-	-	-	-	-	-	-	-	-	-
CO5	3	3	3	-	3	-	-	-	-	-	-	-	-	-	-
Average	3	3	2.4	-	2	-	-	-	-	-	-	-	-	-	-

1 – Weakly Mapped (Low) **2 – Moderately Mapped (Medium)**

3 – Strongly Mapped (High) **"_" means there is no correlation**

Syllabus

Unit 0: Motivation **01 Lecture Hours**

Why study this course- Relevance and Significance?

Unit I: Numerical Methods and Optimization **12 Lecture Hours**

Bisection and Newton-Raphson methods, Gauss Elimination and Gauss-Seidel methods, Finite difference operators, Interpolation with equal and unequal intervals, Numerical differentiation, and integration, Numerical solution of ODEs: Picard's method, Euler's method, Runge-Kutta fourth order method. Introduction to optimization, The Simplex method, Duality, Lagrange multipliers, Convex sets and functions, Elements of Gradient search algorithms: Steepest descent, Newton and Jacobi algorithms, Least Squares method, Application: The Markowitz Model, and Overview of constrained optimization, Hill climbing, Single variable search. Recap of Unit I.

Unit II: Infinite Series and Introduction to Complex Analysis **12 Lecture Hours**

Sequence and series, Convergence tests: p-series, Comparison, Ratio and root test, Alternating series. Complex number system, Euler's formula, Functions of a complex variable, Hyperbolic functions, Limit and Continuity, Derivative and Analytic functions, Holomorphic functions, Cauchy-Riemann equations, Harmonic functions, Line integral and independence of path, Cauchy's theorem, Cauchy's integral formula, Zeros and singularities of a function, Power series: Taylor's and Laurent's series. Some applications. Recap of Unit II.

Unit III: Introduction to Special Functions **04 Lecture Hours**

Introduction to Power series method, Legendre's equation and Legendre polynomials, Bessel's equation and Bessel functions. Application of Bessel functions: CV Raman's model of Indian drums. Recap of Unit III.

Unit IV: Integral Transforms **08 Lecture Hours**

Laplace Transform and its properties, Shifting Theorems, Laplace Transform of derivatives, integrals, and periodic functions, Heaviside and Dirac Delta Functions. Inverse Laplace transforms, Convolution, Solutions of differential equations using Laplace transforms. Fourier series and applications, Dirichlet's condition, Fourier Transforms, Fourier sine and cosine transforms, Properties of Fourier Transforms, Fast Fourier Transform, Inverse Fourier transforms. Recap of Unit IV.

Unit V: Introduction to PDEs and Applications **08 Lecture Hours**

Introduction to Partial differential equations (PDE) and real-world applications, Classification of PDEs: Elliptic, Hyperbolic, Parabolic, Solution of homogeneous and non-homogeneous linear PDEs, Method of separation of variables using Fourier series, Solution of Heat conduction or Diffusion equation, Connection between diffusion and randomness, Wave Equation, Laplace Equation, and Poisson Equation. Some applications: Air pollution, Traffic model. Recap of Unit V.

Total lecture Hours 45

References*

Textbooks	1. E. Kreyszig, "Advanced Engineering Mathematics", 9th Edition, Wiley Publications, 2016. 2. G. James, and P. P. Dyke, "Advanced Modern Engineering Mathematics", 5th Edition, Pearson Education. 2018. 3. J. P. Corriou, "Numerical Methods and Optimization: Theory and Practice for Engineers", Springer, 2021.
Reference books	1. D. G. Zill, and P. D. Shanahan, "Complex analysis", 3rd Edition, Jones & Bartlett Learning, 2015. 2. W. A. Strauss, "Partial Differential Equations - An Introduction", 2nd Edition, John Wiley & Sons Inc., 2008. 3. L. Burstein, "PDE Toolbox Primer for Engineering Applications with MATLAB® Basics", CRC Press, 2022.

	4. I. Goodfellow,Y. Bengio, and A. Courville, "Deep learning", MIT Press., 2016. (Sections 4.3, 4.4, and 4.5) 5. S. Chandra, Jayadeva, and A. Mehra, "Numerical optimization with applications", Narosa, 2009.
Web Resources	1. Gaudet, S., Gauthier, C., & Léger, S. (2006). The evolution of harmonic Indian musical drums: A mathematical perspective. Journal of sound and vibration, 291(1-2), 388-394. 2. MIT OCW on Complex Variables with Applications. https://ocw.mit.edu/courses/18-04-complex-variables-with-applications-spring-2018/ 3. Balakrishnan, V., Selected Topics in Mathematical Physics https://nptel.ac.in/courses/115106086 . 4. Hsu T.R., Applied Engineering Analysis, John Wiley & Sons. ISBN: 9781119071181, 2017. 5. Neunzert H., Industrial Mathematics and India: A View from the Outside, TMCB, https://www.themathconsortium.in/publication/timcbulletin/vol2/issue1
Journals	
MOOCs, online courses	

Modes of Evaluation: Quiz/Assignment/ presentation/ extempore/ Written Examination etc.

Examination Scheme

Components	IA	MID SEM	End Sem	Total
Weightage (%)	50	20	30	100